Points of intersection

1

a 2 | b 1 | c 0 | d 1

e 0 | f 2 | g 1 | h 1

i 0 | j 0

2 $(3,4),(-2,9)$

3 Minimum value of $k=1$

4

a $4x^2-5x+14=0$

b -199

c No, because the discriminant is negative which means the quadratic has no solution.

5 5

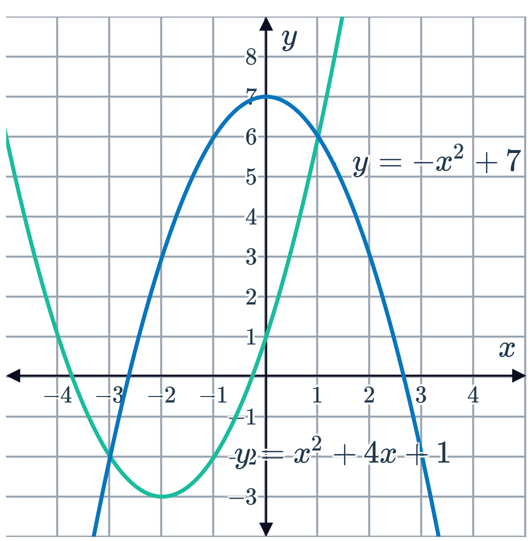
6 $(-5,-67)$

7 $k=7$

8

a

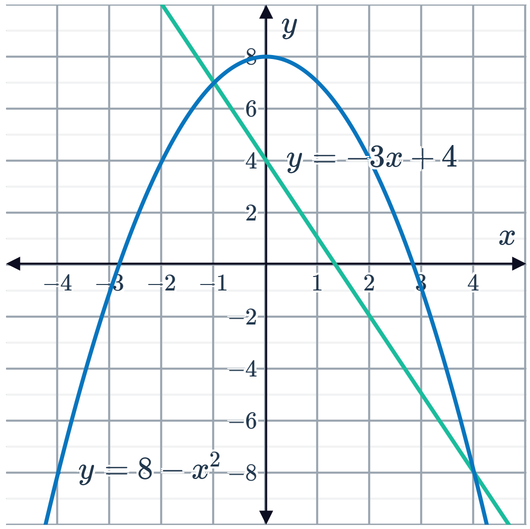
i



ii $(1,6),(-3,-2)$

b

i



ii $(-1, 7), (4, -8)$

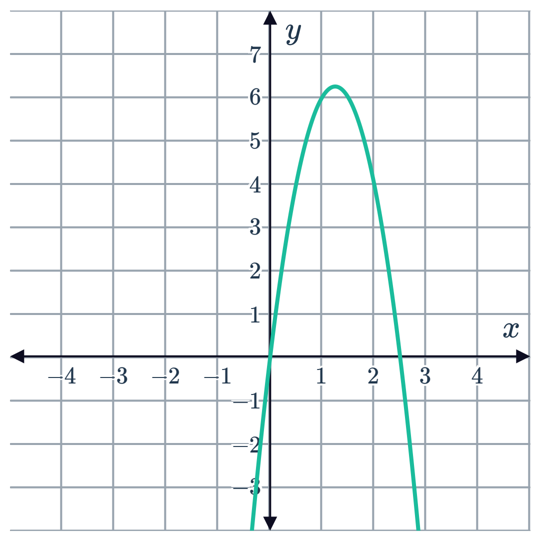
9

a $x = 0, \dfrac{5}{2}$

b $x = \dfrac{5}{4}$

c $\left(\dfrac{5}{4}, \dfrac{25}{4}\right)$

d



e $c = 6.25$

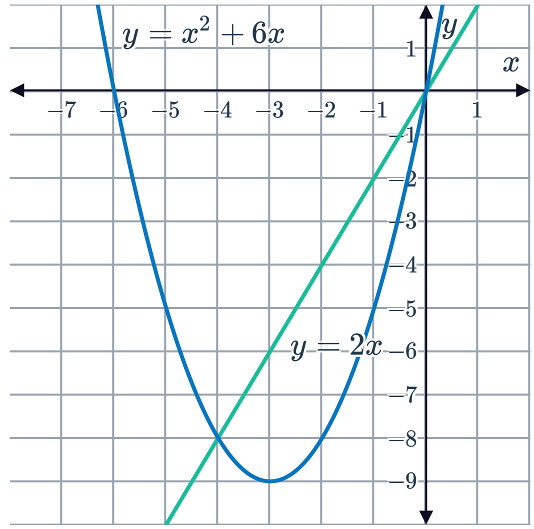
f $c < \dfrac{25}{4}$

10

a

x	-4	-3	-2
$y=x^2+6x$	-8	-9	-8
$y=2x$	-8	-6	-4

b



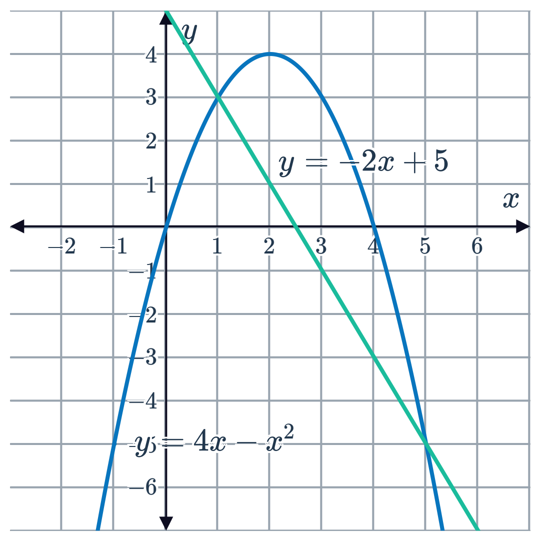
c $(0,0),(-4,-8)$

11

a

x	1	2	3
$y=4x-x^2$	3	4	3
$y=-2x+5$	3	1	-1

b



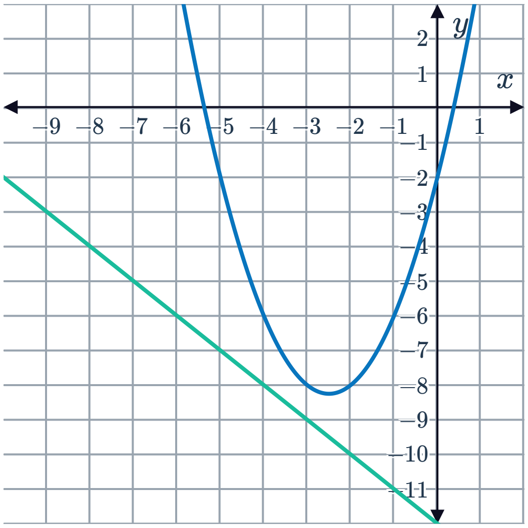
c $(1,3),(5,-5)$

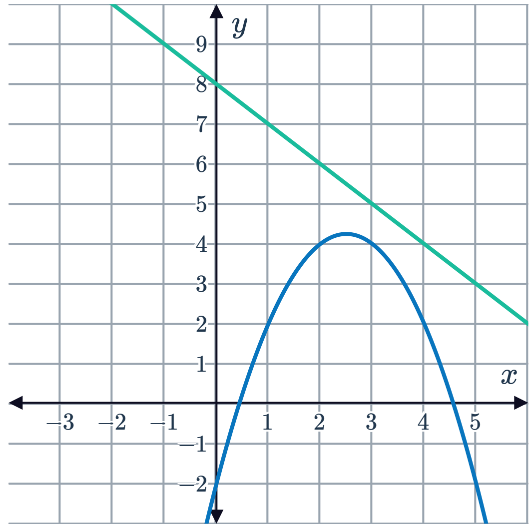
12

a No

b $(-2,7)$

13 Example answers:

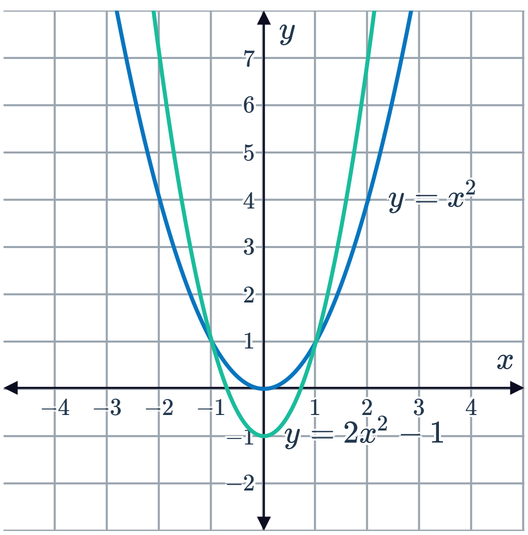




14

a

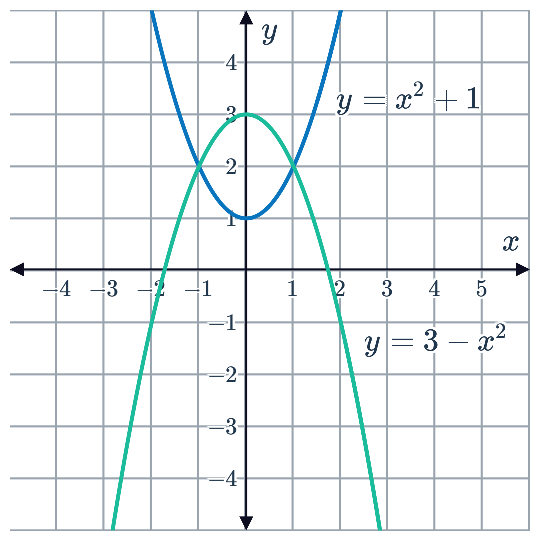
i



ii $(-1, 1), (1, 1)$

b

i



ii $(-1, 2), (1, 2)$

Quadratic systems of equations

15

a

i $x = \pm 2$ ii $(-2, 0), (2, 0)$

b

i $x = \pm 2$ ii $(-2, 4), (2, 4)$

c

i $x = 0$ ii $(0, 0)$

d

i $x = 3, -5$ ii $(3, 25), (-5, 9)$

e

i $x = 7$ ii $(7, 8)$

f

i $x = 5, 4$ ii $(5, 54), (4, 38)$

g

i $x = \frac{1}{4}$ ii $\left(\frac{1}{4}, -\frac{109}{16}\right)$

h

i $x = \frac{21}{2}$ ii $\left(\frac{21}{2}, \frac{5}{4}\right)$

i

i $x = 2$ ii $(2, 6)$

j

i $x = \pm\sqrt{10}$ ii $\left(-\sqrt{10}, -8\right), \left(\sqrt{10}, -8\right)$

16

a $(0, 0), (-7, 49)$ b $(3, 18), (-7, 28)$ c $(3, -7)$ d $(0, 0), (1, 16)$

e $(7, -7), (1, -7)$ f $(-4, 42)$ g $(2, 2), (-3, -3)$ h $(3, 0), \left(\frac{7}{2}, \frac{1}{4}\right)$

i $(2, 1), (-2, 1)$

17

a $x = \frac{1+\sqrt{17}}{2}, \frac{1-\sqrt{17}}{2}$ b $\left(\frac{1+\sqrt{17}}{2}, -4\right), \left(\frac{1-\sqrt{17}}{2}, -4\right)$

18 $(0.16, -1.15), (-6.16, 5.17)$

19

a $(-2,3),(2,-9)$

b $(-4,-12),(-3,-12)$

c $(2,-4),(0,0)$

d

i $x=4,-3$ **ii** $(4,0),(-3,21)$

e

i $x=-4,-5$ **ii** $(-4,2),(-5,5)$

20 $y=-x-\frac{21}{4}$

21

a $x=-4\pm2\sqrt{3}$

b

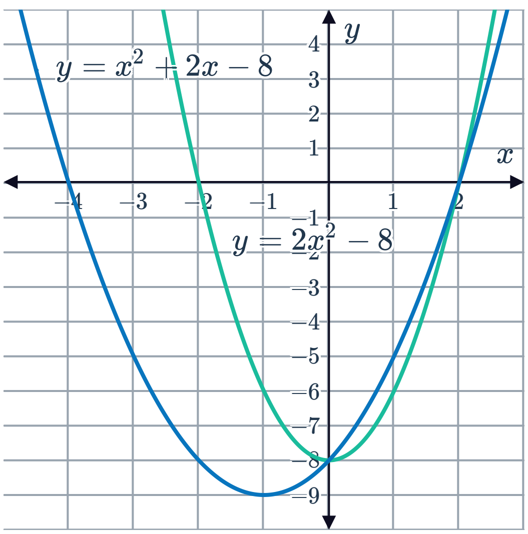
i Yes **ii** Yes **iii** Yes

22

a $c=-8$

b $a=-2$

c

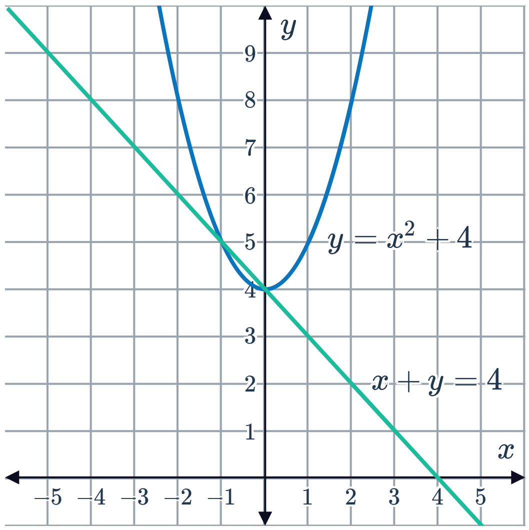


Quadratic equations from systems

23 $3x^2 - 2x + 9 = 0$

24

a



b 2

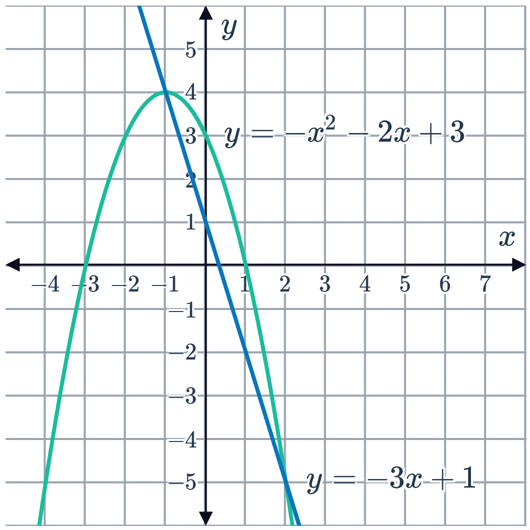
c $(0,4), (-1,5)$

25 $y = 2x + 6$

26

a $-3x + 1$

b

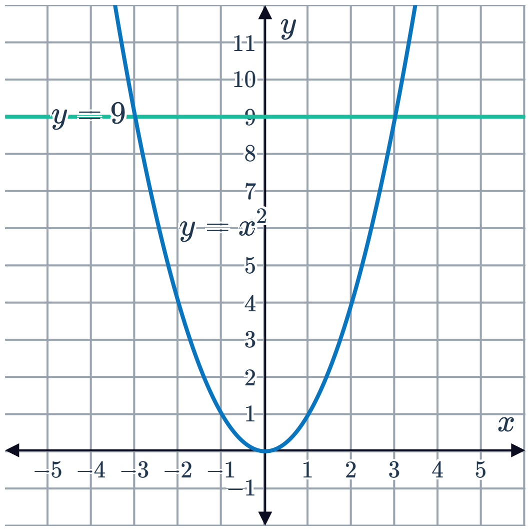


c $x = -1, 2$

27

28

a



b $(-3, 9), (3, 9)$

c $x = \pm 3$

Applications

29 $y = 1, \frac{8}{3}$

30

a $x = 0, x = 6$

b $y = 144$

c Example answer:
The equilibrium quantity occurs when 6 thousand units are being produced, which has a corresponding price of 144 dollars per unit.

31